INCH-POUND

A-A-52050
January 24, 1992

COMMERCIAL ITEM DESCRIPTION

PROTRACTORS, SEMICIRCULAR, PLASTIC, GRADUATED IN MILS, YARDS, METERS, AND INCHES

The General Services Administration has authorized the use of this commercial item description in preference to Military specification MIL-P-20385E.

This description covers semicircular and plastic protractors for use in machine gun and artillery fire control.

Salient characteristics:

Design. The protractors shall be as shown in figures 1 through 5 and as specified herein.

Classification. Protractors shall be of the following types and styles, as specified.

Type I - Graduated in mils, and yards or meters, 1:25,000, and 1:50,000 scale.

Style 1 - 12-inch diameter, graduated in yards.
Style 2 - 12-inch diameter, graduated in meters.
Style 3 - 16-inch diameter, graduated in meters.

Type II - Graduated in mils, yards and inches, or meters and centimeters.

Style 1 - 10-inch diameter, graduated in yards and inches.
Style 2 - 10-inch diameter, graduated in meters and centimeters.

Type I. Type I protractors shall be graduated in mils, and yards or meters with 1:25,000 and 1:50,000 ratio scales. Type I protractors shall be fabricated as shown in figures 1, 2, and 3 for styles 1, 2, and 3, respectively, and as specified herein. The 1:50,000 ratio scale and any markings related thereto shall be made with a permanent red color.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and other data which may improve this document should be sent by letter to: Commander, US Army Belvoir Research, Development and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606.

AMSC N/A

FSC 6675

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Type II. Type II protractors shall be graduated in mils, yards and inches, or meters and centimeters, with 1:10,000, 1:20,000 and 1:40,000 ratio scales. Type II protractors shall be fabricated as shown in figures 4 and 5 for styles 1 and 2, respectively, and as specified herein.

Materials: Material shall be as specified herein.

Plastic: Plastic shall be poly (Vinyl Chloride-Vinyl Acetate) unmodified, colorless and transparent.

Graduations and numerals. Graduations, and numerals shall be molded, impressed, or printed on the underside of the protractor and filled or printed with permanent black color so as to be readable from the upper side. Lettering and numerals shall be a readable, legible size and style. All graduation lines shall be 0.008-inch wide, ±0.002-inch.

Laminations. If the graduations and markings on the lower surface of the protractor are printed, they shall be protected by a transparent plastic cover. The cover sheet shall be permanently affixed by a laminating process which shall not impair the transparency of the protractor or cover sheet, nor distort the markings or graduations. There shall be no evidence of the cover sheet separating from the protractor.

Identification marking. Each protractor shall be identified with the manufacturer's identification.

Workmanship. The protractors shall be free from warpage, cracks, scratches, bubbles, imbedded particles, and other defects that affect serviceability.

Certification of commercial item. The contractor shall certify that the product offered to the Government meets the salient characteristics of this description and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices and is the same product as the manufacturer's current product offered by the contractor in the commercial marketplace for at least one year preceding the solicitation. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this commercial item description are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Quality assurance: The protractors shall be examined for the following defects. Presence of one or more defects shall be caused for rejection.

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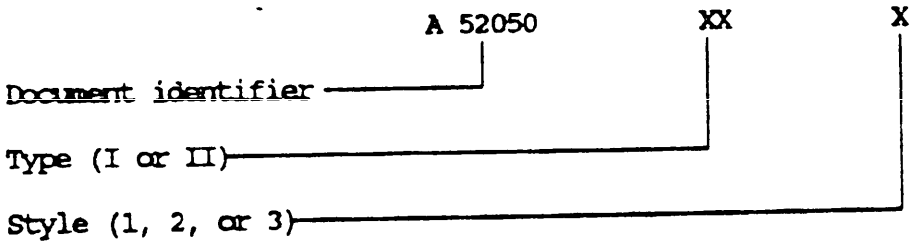
1. Nonconformance to figures.
2. Type not as specified.
3. Style not as specified.
4. Material not as specified.
5. Lettering, numerals, or graduations missing or not easily legible.
6. Numerals or graduation not as specified.
7. Color not as specified.
8. Cover sheet not as specified.
9. Identification marking missing, incomplete, or illegible.
10. Workmanship not as specified.
11. Packing not as specified.

Regulatory requirements. In accordance with Federal Acquisition Regulation, section 23.403, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing the contractor's employees to undue hazards from the recovered materials.

Preservation, packaging, packing, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

CID based part identification numbers. The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

Part or identifying number (PIN). The part or identifying number for protractors, semicircular, (Mils, Yards, Meters, and Inches) shall be as follows:



Notes. The procuring agency should specify the preferred options permitted herein and include the following information in procurement documents:

1. Title, number, and date of this item description.
2. When the commercial item description is used for acquisition, the commercial item certification clause must appear in the solicitation.
3. The ordering activity shall specify type and style required.

FED-STD-376 is available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

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The following are ~~suggested~~ sources of supply for this item description (Competition is not limited to these ~~sources~~.):

1. C-Thru Ruler Co., P.O. Box 356, 6 Britton Drive, Beltsville, MD 06002-3617.
2. Lutz Superdyne Co., Inc., 815 Broad Avenue, Ridgefield, N. J. 07657.

The parparing activity for this item description is the US Army Belvoir Research Development and Engineering Center, ATIN: STRBE-TSE, Fort Belvoir, VA 22060-5606.

MILITARY INTEREST:

Custodians
Army - ME
Air Force - 99

Review activity
DLA - GS

User activities
Army - AR
Navy - MC

CIVIL AGENCY COORDINATING ACTIVITIES.

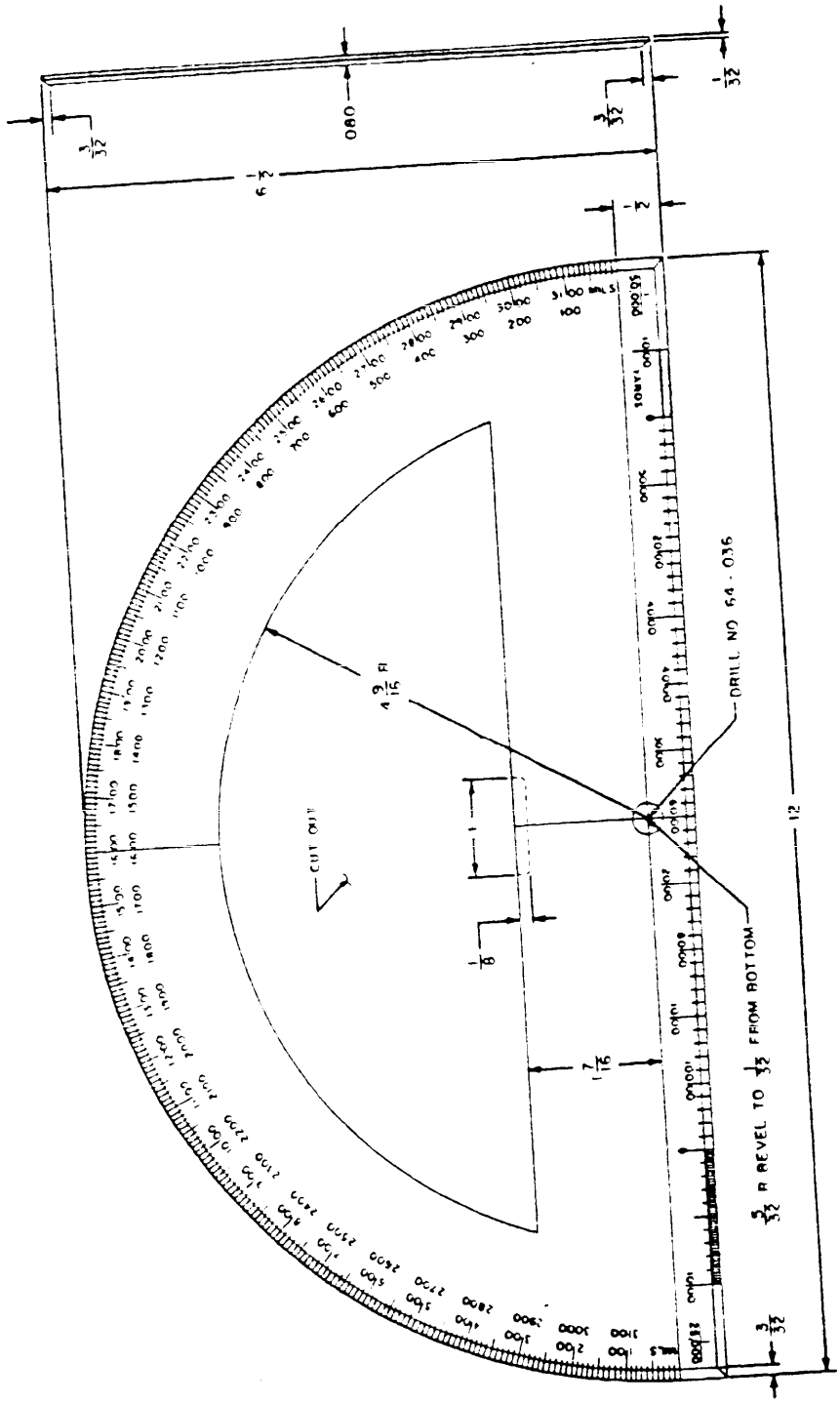
GSA-FOGC

PREPARING ACTIVITY:

Army - ME

Project 6675-0897

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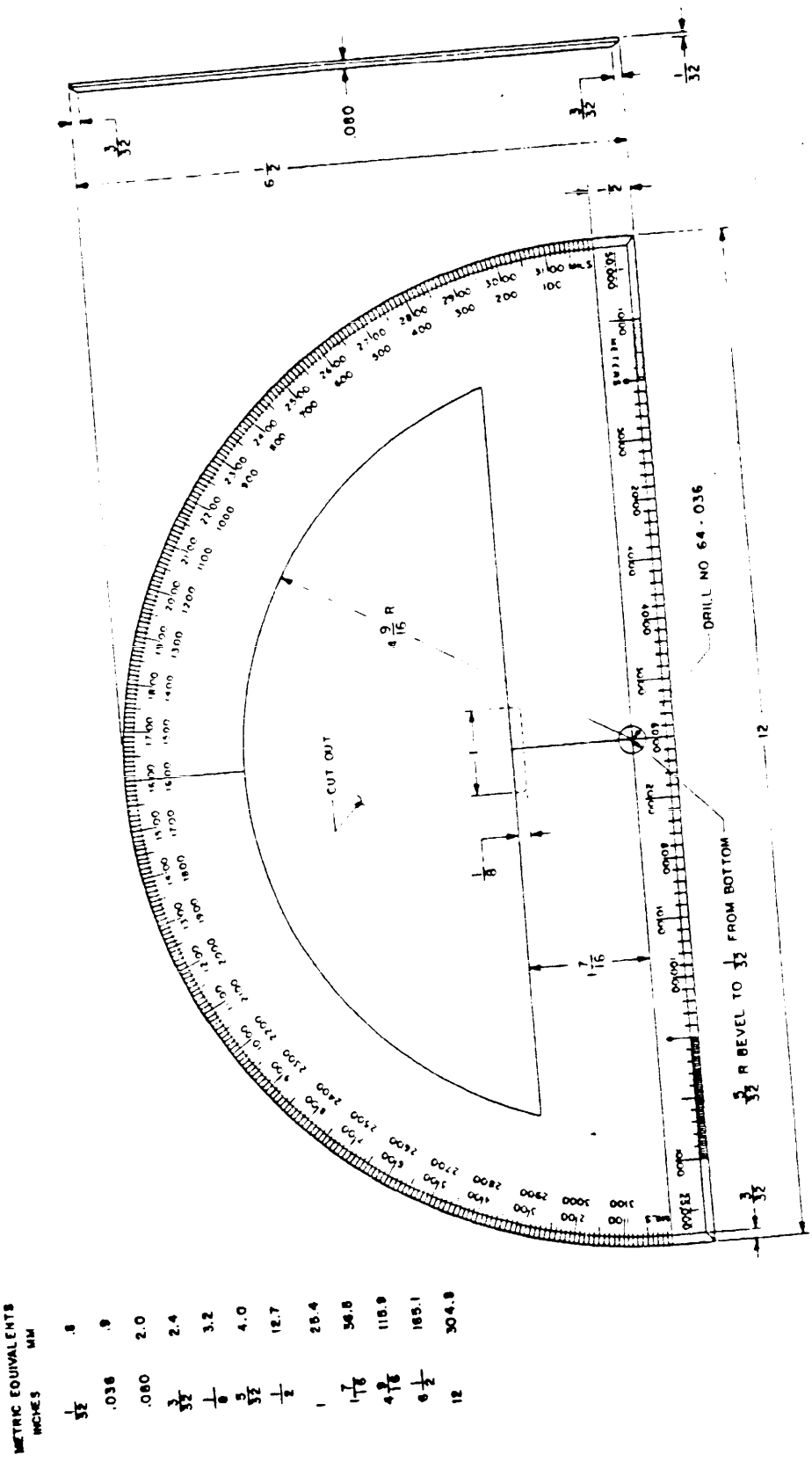
METRIC EQUIVALENTS

INCHES	MM
1/32	.8
.036	.9
.080	2.0
3/32	2.4
1/8	3.2
5/32	4.0
1/2	12.7
1	25.4
1 7/16	36.5
4 9/16	115.9
6 1/2	165.1
12	304.8

FIGURE 1. Protractor, scales 1/25,000 and 1/50,000, 12-inch diameter, yards.

X-1025A

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METRIC EQUIVALENTS

INCHES	MM
$\frac{1}{32}$	.8
.035	.9
.080	2.0
$\frac{3}{32}$	2.4
$\frac{1}{8}$	3.2
$\frac{5}{32}$	4.0
$\frac{1}{2}$	12.7
1	25.4
$1\frac{7}{16}$	36.5
$4\frac{9}{16}$	116.3
$6\frac{1}{2}$	165.1
12	304.8

FIGURE 2. Protractor, scales 1/25,000 and 1/50,000, 12-inch, diameter, meters.

X-3233

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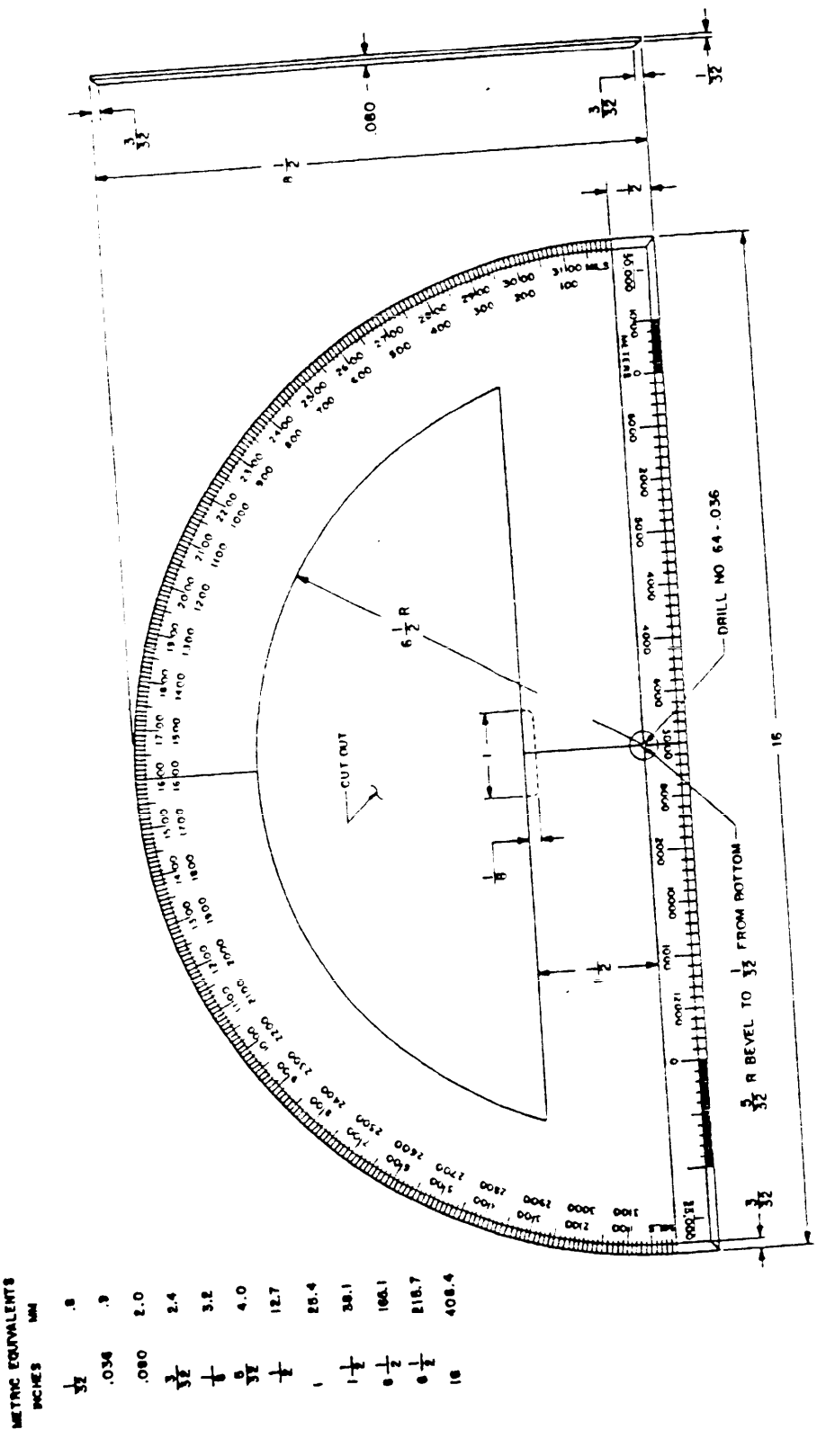


FIGURE 3. Protractor, scales 1/25,000 and 1/50,000, 16-inch diameter, meters. X-1026A

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X-1027A

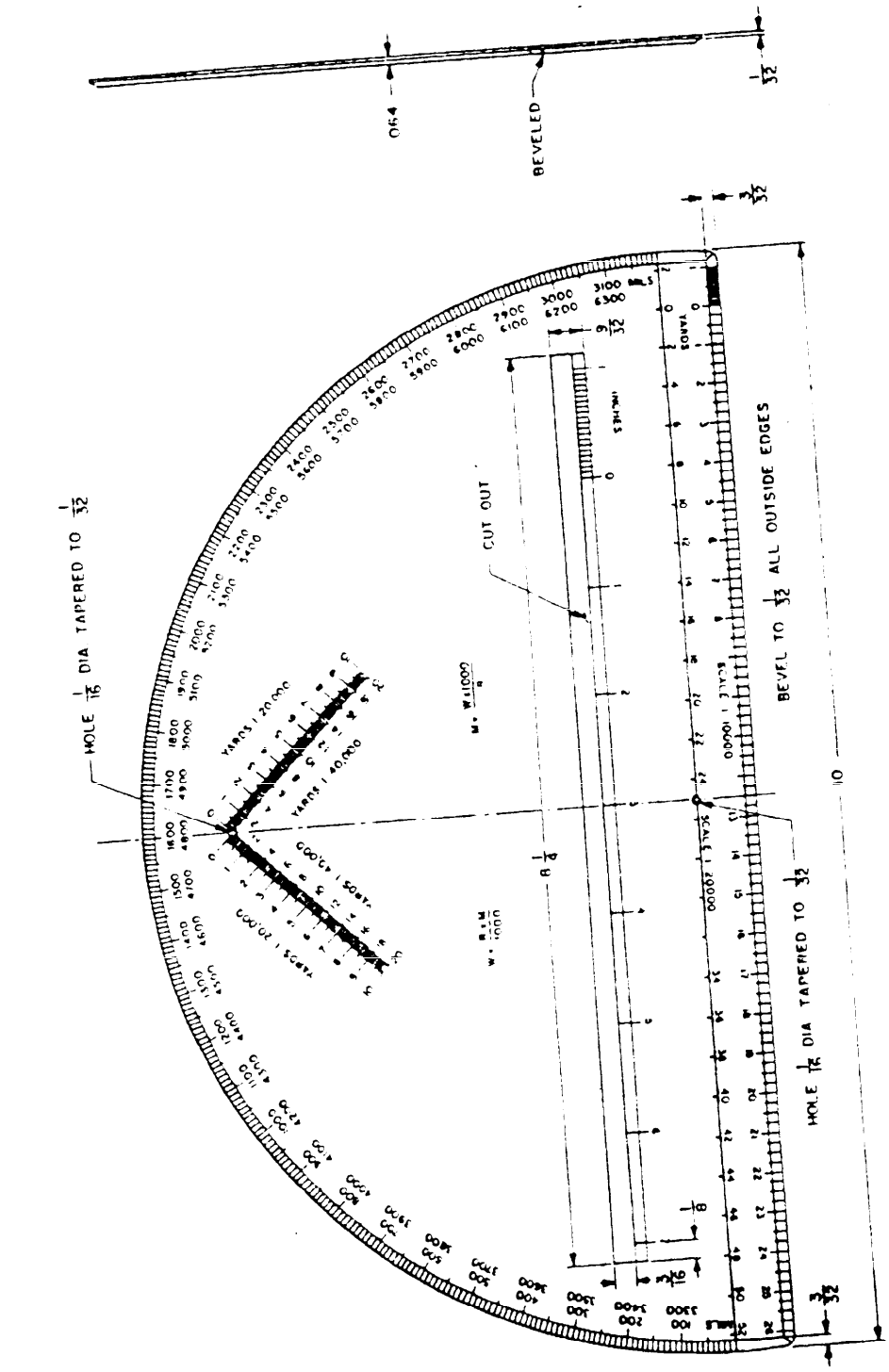
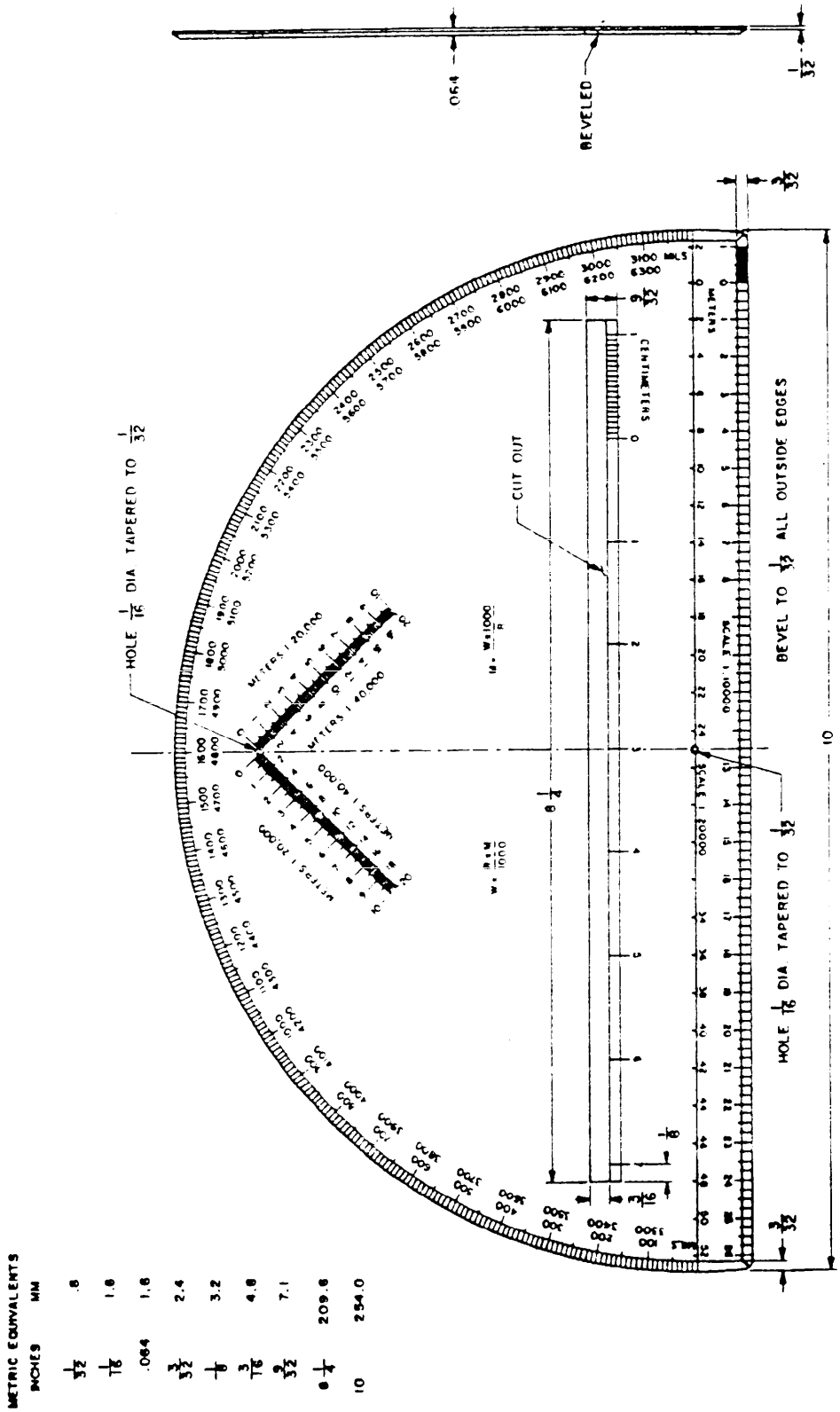


FIGURE 4. Protractor, scales 1/10,000, 1/20,000 and 1/40,000, 10-inch diameter, yards and inches.

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X-3234

FIGURE 5. Protractor, scales 1/10,000, 1/20,000 and 1/40,000, 10-inch diameter, meters and centimeters.

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